SK hynix

SC26 Chicago, IL hpc unites.

Custom-HBM Array

LLM Decode Accelerator

Paper

e-Brochure

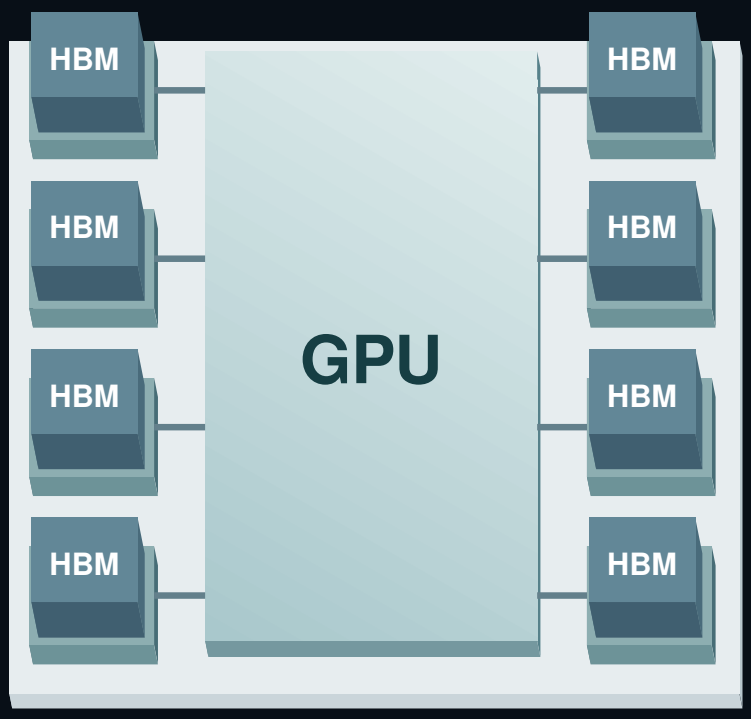
The decoding bottleneck

Memory-bound LLM decoding leaves traditional GPU compute underutilized, while HBM bandwidth is difficult to scale.

Scalable cHBM chiplets

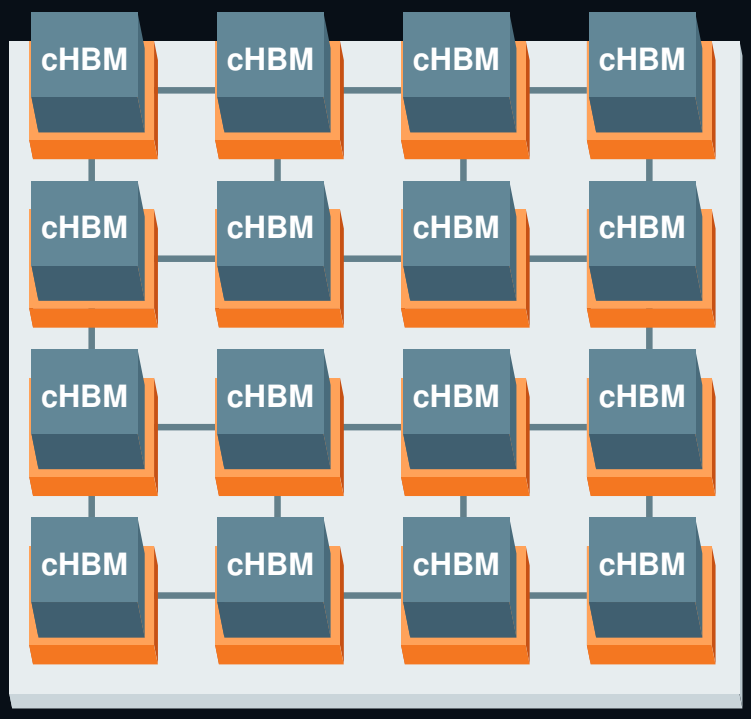
Each cHBM chiplet integrates an NPU into its logic die for LLM decoding. The array scales memory bandwidth to match workload needs.

Conventional GPU + HBM



High memory use, low compute utilization

4 × 4 cHBM Array

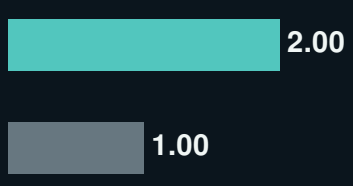


Hybrid tensor and pipeline parallelism

Performance

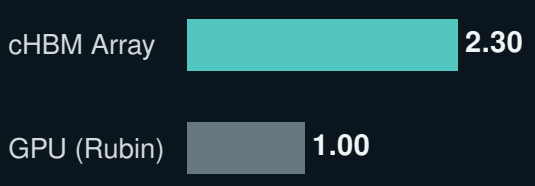
2.00× Memory bandwidth

Memory bandwidth



UP TO **2.30×** Decode throughput

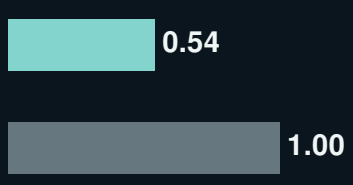
Tokens per second



Energy efficiency

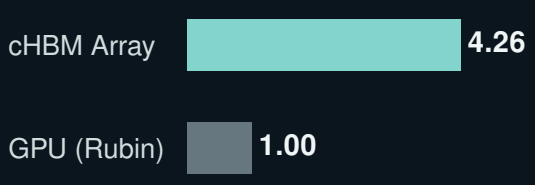
-46% Power consumption

Power consumption



UP TO **4.26×** Performance per watt

Tokens per second / watt



Relative to GPU (Rubin). Chart values normalize GPU (Rubin) to 1.00.